



VIT-AP
UNIVERSITY

DIGITAL LOGIC DESIGN

(Course Code: ECE 1003)

Module-1:Lecture-8

Boolean Algebra

CONTENTS

Module-1 (Part-2)

- ❖ Logic gates and universal gates
- ❖ Tri-state logic
- ❖ **Boolean algebra**
- ❖ SOP and POS minimization
- ❖ K-maps upto 4-variables,
- ❖ Don't care logic

CONTENTS

Lecture-8

❖ Boolean Algebra

BOOKS

Textbooks

1. M.Morris Mano, Michael D Ciletti, **Digital Design**, 5th edition, Pearson Publishers, 2013.
2. R.P. Jain, “**Modern Digital Electronics**”, 4th edition, TMH.

References

1. M.Morris Mano, Charles R. Kime, Tom Martin, **Logic and Computer Design Fundamentals**, 4th edition, Pearson Publishers.
2. C. H. Roth and L. L. Kinney, **Fundamentals of Logic Design**, 5th edition, Cengage Publishers.

Boolean Algebra

BOOLEAN ALGEBRA

- ✓ Developed by English Mathematician George Boole in between 1815 - 1864.
- ✓ It is described as **an algebra of logic** or an algebra of two values i.e **True or False**.
- ✓ The term logic means a statement having binary decisions i.e **True/Yes or False/No**.
- ✓ It is used to perform the logical operations in digital computer.
- ✓ In digital computer **True** represent by '**1**' (**high voltage**) and **False** represent by '**0**' (**low voltage**).
- ✓ Logical operations are performed by logical operators. The fundamental logical operators are:
 1. **AND (conjunction)**
 2. **OR (disjunction)**
 3. **NOT (negation/complement)**

BOOLEAN ALGEBRA

Truth Table

- ✓ **Truth table** is a table that contains all **possible values of logical variables/statements** in a Boolean expression.
- ✓ No. of possible combinations = 2^n , where n =number of variables used in a Boolean expression.
- ✓ The truth table for $XY + Z$ is as follows:

Dec	X	Y	Z	X.Y	X.Y+Z
0	0	0	0	0	0
1	0	0	1	0	1
2	0	1	0	0	0
3	0	1	1	0	1
4	1	0	0	0	0
5	1	0	1	0	1
6	1	1	0	1	1
7	1	1	1	1	1

BOOLEAN ALGEBRA

General Definition

- ❑ Set of elements, a set of operators, and a number postulates
 - ❑ $a * b = c$, $*$ is a binary operator if it specifies a rule for finding c from the pair (a, b) and also if $a, b, c \in S$.
 - ❑ $*$ is not a binary operator if $a, b \in S$, if $c \notin S$.
 - ❑ Common postulates
- ✓ There are two different ways to prove different theorems in Boolean Algebra
1. Truth Table Method
 2. Algebraic Method

BOOLEAN ALGEBRA

Postulates

1) Complementation Law

✓ Like a NOT operator

$$\bar{0} = 1, \quad \text{if } A = 0, \text{ then } \bar{A} = 1$$

$$\bar{1} = 0, \quad \text{if } A = 1, \text{ then } \bar{A} = 0$$

$$\text{Postulate-1: } x \cdot \bar{x} = 0$$

$$\text{Postulate-2: } x + \bar{x} = 1$$

Truth Table

x	$\bar{x}$	$x \cdot \bar{x}$	$x + \bar{x}$
0	1	0	1
1	0	0	1

BOOLEAN ALGEBRA

Postulates

2) Identity Property

Postulate-3: $x \cdot 1 = x$

Postulate-4: $x + 0 = x$

Truth Table

x	$x \cdot 1$	$x + 0$
0	0	0
1	1	1

BOOLEAN ALGEBRA

Postulates

3) Commutative Property

Postulate-5: $x + y = y + x$

Postulate-6: $x \cdot y = y \cdot x$

Truth Table
(P-5)

x	y	$x + y$	$y + x$
0	0	0	0
0	1	1	1
1	0	1	1
1	1	1	1

Truth Table
(P-6)

x	y	$x \cdot y$	$y \cdot x$
0	0	0	0
0	1	0	0
1	0	0	0
1	1	1	1

BOOLEAN ALGEBRA

Postulates

4) Associative Property

$$\text{Postulate-7: } x + (y + z) = (x + y) + z$$

$$\text{Postulate-8: } x \cdot (y \cdot z) = (x \cdot y) \cdot z$$

Note: To simplify notation, the AND (.) operator is frequently omitted. When two elements are written next to each other, the AND (.) operator is implied. Simply, we can write

$$x(yz) = (xy)z$$

BOOLEAN ALGEBRA

Postulates: Associative Property (P-7): $x + (y + z) = (x + y) + z$

x	y	z	$y + z$	$x + y$	$x + (y + z)$	$(x + y) + z$
0	0	0	0	0	0	0
0	0	1	1	0	1	1
0	1	0	1	1	1	1
0	1	1	1	1	1	1
1	0	0	0	1	1	1
1	0	1	1	1	1	1
1	1	0	1	1	1	1
1	1	1	1	1	1	1

BOOLEAN ALGEBRA

Postulates: Associative Property (P-8): $x \cdot (y \cdot z) = (x \cdot y) \cdot z$

x	y	z	$y \cdot z$	$x \cdot y$	$x \cdot (y \cdot z)$	$(x \cdot y) \cdot z$
0	0	0	0	0	0	0
0	0	1	0	0	0	0
0	1	0	0	0	0	0
0	1	1	1	0	0	0
1	0	0	0	0	0	0
1	0	1	0	0	0	0
1	1	0	0	1	0	0
1	1	1	1	1	1	1

BOOLEAN ALGEBRA

Postulates

5) Distributive Property

Postulate-9: $x(y + z) = xy + xz$

Postulate-10: $x + yz = (x + y)(x + z)$

✓ Postulates 9 and 10 are dual to each other.

BOOLEAN ALGEBRA

Postulates: Distributive Property (P-9): $x(y + z) = xy + xz$

x	y	z	$y + z$	xy	xz	$x(y + z)$	$xy + xz$
0	0	0	0	0	0	0	0
0	0	1	1	0	0	0	0
0	1	0	1	0	0	0	0
0	1	1	1	0	0	0	0
1	0	0	0	0	0	0	0
1	0	1	1	0	1	1	1
1	1	0	1	1	0	1	1
1	1	1	1	1	1	1	1

BOOLEAN ALGEBRA

Postulates: Distributive Property (P-10): $x + yz = (x + y)(x + z)$

x	y	z	yz	$x + y$	$x + z$	$x + yz$	$(x + y)(x + z)$
0	0	0	0	0	0	0	0
0	0	1	0	0	1	0	0
0	1	0	0	1	1	0	0
0	1	1	1	1	1	1	1
1	0	0	0	1	1	1	1
1	0	1	0	1	1	1	1
1	1	0	0	1	1	1	1
1	1	1	1	1	1	1	1

BOOLEAN ALGEBRA

Duality:

- ✓ The principle of duality is an important concept. This says that if an expression is valid in Boolean algebra, the dual of that expression is also valid.
- ✓ To form the dual of an expression, replace all + operators with . operators, all . operators with + operators, all ones with zeros, and all zeros with ones.
- ✓ Form the dual of the expression

$$a + (bc) = (a + b)(a + c)$$

- ✓ Following the replacement rules...

$$a(b + c) = ab + ac$$

- ✓ Take care not to alter the location of the parentheses if they are present.

BOOLEAN ALGEBRA

Theorems

1) Idempotent Law

Theorem-1: $x \cdot x = x$

Theorem-2: $x + x = x$

Truth Table

x	$x \cdot x$	$x + x$
0	0	0
1	1	1

BOOLEAN ALGEBRA

Theorems

2) AND and OR Law

Theorem-3: $x \cdot 0 = 0$

Theorem-4: $x + 1 = 1$

Truth Table

x	$x \cdot 0$	$x + 1$
0	0	1
1	0	1

BOOLEAN ALGEBRA

Theorems

3) Absorption Law

Theorem-5: $x + xy = x$

Theorem-6: $x(x + y) = x$

Truth Table
(T-5)

x	y	xy	$x + xy$
0	0	0	0
0	1	0	0
1	0	0	1
1	1	1	1

Truth Table
(T-6)

x	y	$x + y$	$x(x + y)$
0	0	0	0
0	1	1	0
1	0	1	1
1	1	1	1

✓ Theorems 9 and 10 are dual to each other.

BOOLEAN ALGEBRA

Theorems

4) De-Morgan's Law/ Complement Law

Theorem-7: $\overline{x + y} = \bar{x} \cdot \bar{y}$

Theorem-8: $\overline{xy} = \bar{x} + \bar{y}$

Duality	Complement
$\cdot \rightarrow +$	$\cdot \rightarrow +$
$+ \rightarrow \cdot$	$+ \rightarrow \cdot$
$\bar{0} \rightarrow 1$	$\bar{0} \rightarrow 1$
$\bar{1} \rightarrow 0$	$\bar{1} \rightarrow 0$
	$F \rightarrow F'$

BOOLEAN ALGEBRA

Theorems: De-Morgan's Law (T-7): $\overline{x + y} = \bar{x} \cdot \bar{y}$

x	y	$x + y$	$\overline{x + y}$	$\bar{x}$	$\bar{y}$	$\bar{x} \cdot \bar{y}$
0	0	0	1	1	1	1
0	1	1	0	1	0	0
1	0	1	0	0	1	0
1	1	1	0	0	0	0

BOOLEAN ALGEBRA

Theorems: De-Morgan's Law (T-8): $\overline{xy} = \bar{x} + \bar{y}$

x	y	xy	$\overline{xy}$	$\bar{x}$	$\bar{y}$	$\bar{x} + \bar{y}$
0	0	0	1	1	1	1
0	1	0	1	1	0	1
1	0	0	1	0	1	1
1	1	1	0	0	0	0

BOOLEAN ALGEBRA

Theorems

5) Involution Theorem

$$\text{Theorem-9: } (x')' = x$$

Truth Table

x	x'	$(x')'$
0	1	0
1	0	1

BOOLEAN ALGEBRA

Theorems

6) Consensus Law

Theorem-10: $xy + \bar{x}z + yz = xy + \bar{x}z$

Theorem-11: $(x + y)(\bar{x} + z)(y + z) = (x + y)(\bar{x} + z)$

✓ Theorems 10 and 11 are dual to each other.

BOOLEAN ALGEBRA

Theorems: Consensus Law (T-10): $xy + \bar{x}z + yz = xy + \bar{x}z$

x	y	z	$\bar{x}$	xy	$\bar{x}z$	yz	$xy + \bar{x}z + yz$	$xy + \bar{x}z$
0	0	0	1	0	0	0	0	0
0	0	1	1	0	1	0	1	1
0	1	0	1	0	0	0	0	0
0	1	1	1	0	1	1	1	1
1	0	0	0	0	0	0	0	0
1	0	1	0	0	0	0	0	0
1	1	0	0	1	0	0	1	1
1	1	1	0	1	0	1	1	1

BOOLEAN ALGEBRA

Theorems: Consensus Law (T-11): $(x + y)(\bar{x} + z)(y + z) = (x + y)(\bar{x} + z)$

x	y	z	$\bar{x}$	$x + y$	$\bar{x} + z$	$y + z$	$(x + y)(\bar{x} + z)(y + z)$	$(x + y)(\bar{x} + z)$
0	0	0	1	0	1	0	0	0
0	0	1	1	0	1	1	0	0
0	1	0	1	1	1	1	1	1
0	1	1	1	1	1	1	1	1
1	0	0	0	1	0	0	0	0
1	0	1	0	1	1	1	1	1
1	1	0	0	1	0	1	0	0
1	1	1	0	1	1	1	1	1

BOOLEAN ALGEBRA

T1 : Properties of 0

(a) $0 + A = A$

(b) $0 \cdot A = 0$

T2 : Properties of 1

(a) $1 + A = 1$

(b) $1 \cdot A = A$

T3 : Commutative Law

(a) $A + B = B + A$

(b) $A B = B A$

T4 : Associate Law

(a) $(A + B) + C = A + (B + C)$

(b) $(A B) C = A (B C)$

T5 : Distributive Law

(a) $A (B + C) = A B + A C$

(b) $A + (B C) = (A + B) (A + C)$

(c) $A + A'B = A + B$

T6 : Indempotence (Identity) Law

(a) $A + A = A$

(b) $A A = A$

T7 : Absorption (Redundance) Law

(a) $A + A B = A$

(b) $A (A + B) = A$

T8 : Complementary Law

(a) $X + X' = 1$

(b) $X \cdot X' = 0$

T9 : Involution

(a) $X'' = X$

T10 : De Morgan's Theorem

(a) $(X + Y)' = X' \cdot Y'$

(b) $(X \cdot Y)' = X' + Y'$

EXAMPLES

Question: Complement the Boolean function $a(b + z(x + a'))$ and find the dual of the function

$$\begin{aligned}\text{Complement: } (a(b+z(x + a')))' &= a' + (b + z(x + a'))' \\ &= a' + b'(z(x + a'))' \\ &= a' + b'(z' + (x + a'))' \\ &= a' + b'(z' + x'(a'))' \\ &= a' + b'(z' + x'a)\end{aligned}$$

$$\text{Dual: } \quad a(b+z(x + a')) \quad = a + b(z + xa')$$

EXAMPLES

1. $F_1 = x'yz' + x'y'z.$

The dual of F_1 is $(x' + y + z')(x' + y' + z).$

Complement each literal: $(x + y' + z)(x + y + z') = F_1'.$

2. $F_2 = x(y'z' + yz).$

The dual of F_2 is $x + (y' + z')(y + z).$

Complement each literal: $x' + (y + z)(y' + z') = F_2'.$

ASSIGNMENT

Question: Find the complement and dual of the following Boolean function

1. $F = x'z + yz + x(y' + z)$

2. $F = A + B(C + D'(A' + B'))'$

Thank You